

IST 505 Phase 3 Draft

Submission-shaped draft from current repo materials

Phase 3 Draft

IDSS: An Institutionally Adaptive Instructional Decision Support System for Synchronous Online Learning

Working note: This draft is designed to turn the current repo materials into a submission-shaped document for IST 505 Phase 3. It is intentionally conservative about claims that still need literature citations or stronger validation evidence.

1. Problem Statement

In synchronous online learning environments, instructors must continuously interpret multiple streams of partial and fast-changing information, including chat activity, participation patterns, silence, confusion signals, and changes in class energy. Existing dashboards and analytics tools can surface behavioral indicators, but instructors still face a cognitive burden in deciding what those indicators mean and whether intervention is warranted. This creates a situation-awareness problem: instructors may perceive signals without fully comprehending their significance or projecting how engagement is likely to evolve. The proposed artifact addresses this problem by supporting instructor sensemaking and intervention selection in real time, while preserving instructor agency rather than replacing it.

2. DSR Artifact Description

The project contains two tightly related artifacts:

- **Primary artifact: IDSS** An instructional decision-support concept for synchronous online teaching. It aggregates classroom engagement signals, detects patterns, and presents intervention recommendations to instructors.
- **Evaluation artifact: SAGE** A simulation environment used to generate classroom data, test dashboard behavior, and examine instructor responses before real-world deployment.

These artifacts collectively cover multiple DSR artifact types:

DSR artifact type	Project element
Construct	Instructor Response Taxonomy
Model	Situation-awareness pipeline from signal perception to comprehension and projection
Method	Simulation-based formative evaluation using scenario-driven classroom runs
Instantiation	Dashboard prototype, simulation engine, university presets, and professor-response loop

3. What Is Unique Or Novel

The novelty of the project should not be framed as merely adding AI-generated recommendations to an instructor dashboard. A more defensible claim is that the project combines three contributions that are rarely treated together.

First, the artifact is **institutionally adaptive**: the same dashboard and scenario can produce different recommendation patterns depending on institutional context. Second, the design explicitly separates **demographic composition from behavioral modeling**, allowing classroom composition to reflect real enrollment patterns without claiming that race, ethnicity, gender, or nationality directly determine engagement behavior. Third, the project introduces an **Instructor Response Taxonomy** that treats the instructor as an active decision-maker who may ignore, acknowledge, accept, modify, or reject a recommendation. This reframes instructor-AI interaction as a measurable construct rather than a black box.

An additional methodological contribution is that SAGE provides a **simulation-based evaluation environment** for the instructional artifact. This supports formative testing of recommendations, intervention effects, and instructor responses before moving to live classroom settings.

This methodological direction can be situated alongside prior generative-agent work such as Park et al. (2023), which is often summarized through the well-known example of agents in a simulated town spontaneously organizing a Valentine's Day party. In contrast, the present project applies shared-context agent simulation to classroom engagement and instructional decision support.

4. Design Requirements And Principles

The current build implies the following design principles:

ID	Principle	Rationale	Artifact expression
DP1	Institutional context sensitivity	Online classrooms differ structurally across institutions and program formats	University presets with institutional modifiers
DP2	Composition-behavior separation	Demographic realism should not require demographic stereotyping	Demographics describe classroom composition, not direct behavior rules
DP3	Instructor agency preservation	Decision support should advise rather than dictate	Instructor Response Taxonomy and recommendation interface
DP4	Cognitive load reduction	Instructors need aggregated interpretation, not only raw behavioral streams	Dashboard summaries, alerts, and grouped patterns
DP5	Situation awareness escalation	Perception alone is insufficient for intervention	Signals feed into pattern detection and forward-looking recommendations
DP6	Bias-aware evaluation	DSR should surface and reduce bias risks during artifact design	Explicit rejection of demographic-to-behavior mappings
DP7	Risk-reduced formative testing	Early evaluation should be possible without immediate classroom deployment	SAGE as simulation sandbox for scenario testing

Kernel Theory To Feature Mapping

One of the main revision needs for Phase 3 is to show more explicitly how kernel theories inform concrete design features. The current project supports the following mapping:

Kernel theory or literature base	Design implication	Current feature or artifact element
Endsley situation awareness	Move beyond raw signal display toward interpretation and projection	Signal collection, pattern detection, and intervention recommendations

Kernel theory or literature base	Design implication	Current feature or artifact element
Sweller cognitive load theory	Reduce extraneous cognitive burden on instructors	Dashboard aggregation, alerts, and summarized recommendation interface
Advising-mode literature in learning analytics	Recommendations should support action, not just mirror data	Intervention suggestions rather than raw metrics alone
Instructor heterogeneity literature	Different instructors and contexts require adaptive support	Multiple professor styles and institutional-context presets
Bias-aware and equity-oriented design	Avoid encoding demographic assumptions into learner behavior	Composition-behavior separation and rejection of demographic behavior rules
Design science evaluation literature	Use iterative formative evaluation before naturalistic deployment	Scenario-driven simulation and staged evaluation roadmap

5. Form And Functions

The current proof-of-concept includes a dashboard prototype and a simulation engine that work together as a single demonstration environment.

Key functional elements include:

- A dashboard that displays classroom engagement patterns, alerts, and recommendations
- A simulation engine that generates minute-by-minute classroom states under repeatable scenarios
- University presets that shift classroom conditions across institutional contexts
- A professor-response loop that records how recommendations are accepted, modified, or rejected
- Multiple evaluation scenarios, including energy decay, equity imbalance, confusion clusters, and mixed full-session scenarios

The interface can be described in terms of a situation-awareness pipeline:

Layer	Function	Example implementation
Perception	Collect classroom signals	Chat, speaking, reactions, silence, camera state, participation
Comprehension	Detect meaningful patterns	Confusion cluster, energy decay, speaking imbalance
Projection	Recommend possible interventions	Poll, breakout, pace change, clarification

For the final Phase 3 submission, this section should include screenshots or diagrams of:

- Main dashboard layout
- Recommendation panel
- Instructor response area
- University preset comparison view
- Simulation-to-dashboard flow

6. Proof Of Concept Demonstration

The current proof of concept is implemented as a working prototype composed of:

- A Python-based simulator for synthetic classroom sessions
- A browser-based dashboard interface
- Scenario controls and university presets
- Rule-based and LLM-assisted simulation modes
- A prototype live-integration path for Zoom-style event ingestion

The prototype already supports:

- Scenario-driven runs with seeded reproducibility in rule-based mode
- LLM-powered student chat in server mode
- A professor-decision loop that can react to dashboard recommendations
- Exportable session traces and comparison scenarios

For the Phase 3 writeup, the proof-of-concept argument should emphasize that the project is already capable of demonstrating the artifact's intended workflow end to end:

1. A classroom state is generated or ingested
2. The dashboard interprets the state and identifies patterns
3. The system presents interventions to the instructor

4. The instructor response is recorded
5. The next classroom state reflects that intervention context

7. Evaluation Approach And Artifact Impact

The evaluation story should distinguish clearly between current formative evidence and future summative evidence.

Current formative evidence

- Design decision traces, especially the demographic design rationale
- Scenario-based simulation runs across multiple classroom conditions
- Comparison across institutional presets
- Observation of recommendation patterns and instructor-response behavior in the prototype

Current impact claims we can support

- The artifact can detect and surface classroom patterns in a coherent dashboard interface
- The artifact can vary its outputs across institutional contexts
- The artifact can represent instructor response as a meaningful part of the evaluation loop
- The project demonstrates technical feasibility for simulation-based formative evaluation

Future impact claims that need further study

- Whether instructors using the artifact make better decisions than those not using it
- Whether the dashboard reduces cognitive load in real teaching contexts
- Whether the recommendations improve student engagement outcomes in live classrooms
- Whether the live integration path works reliably in real deployment environments

8. Competing-Systems Positioning Scaffold

The literature and related-systems section should avoid claiming that current tools are purely descriptive. A safer framing is that some current tools already provide analytics summaries and, in some cases, recommendation-like support. The differentiation argument should therefore focus on what this project combines that those systems typically do not foreground: institutional adaptation, explicit separation of demographic composition from behavior modeling, and a construct for measuring instructor response to AI support.

A useful structure for the final paper is:

1. Existing systems can surface behavioral indicators and sometimes suggest actions.

2. Those systems are not the central novelty target for this paper.
3. The present artifact is differentiated by context sensitivity, bias-aware modeling choices, and instructor agency as a measurable construct.

Suggested Closing Paragraph

This project contributes an institutionally adaptive instructional decision-support concept and a simulation-based evaluation environment for synchronous online learning. Its main contribution is not simply the addition of AI recommendations, but the combination of context-sensitive recommendation behavior, bias-aware classroom modeling, and an instructor response construct that preserves agency while enabling evaluation. In its current form, the project offers a credible proof of concept and a strong basis for formative design science evaluation, while also identifying a clear path toward more naturalistic future validation.

What Still Needs To Be Added Before Submission

- Literature citations and references formatted for the course submission
- Explicit kernel-theory-to-feature mapping in the final prose
- Screenshots, diagrams, or UI sketches
- A short competing-systems paragraph with citation support
- Final agreement on whether the paper centers IDSS terminology, SAGE terminology, or both

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